Analyze the Effectiveness of Using ECMO for Patients

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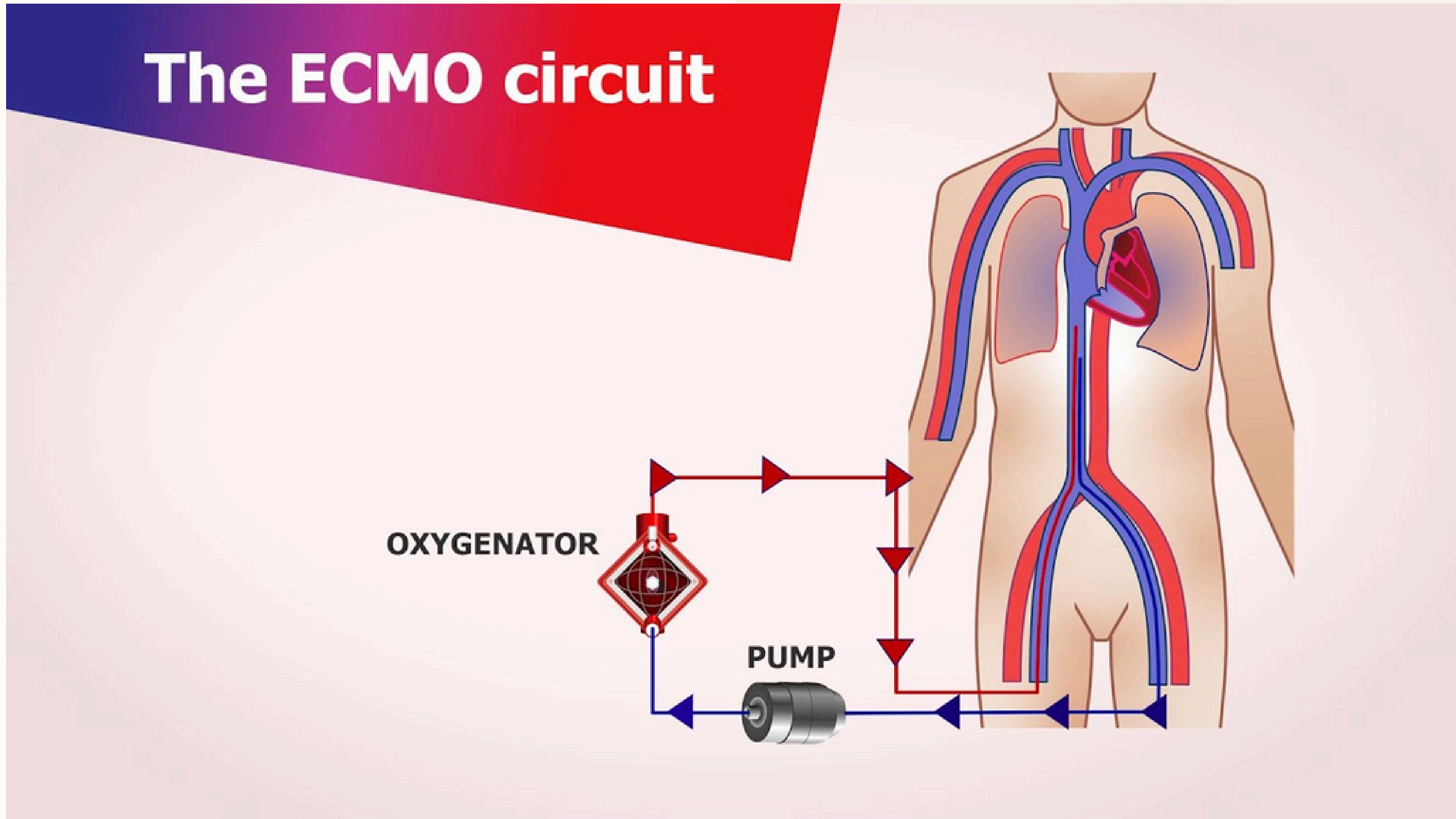
List of Contents

Introduction

ECMO

ECMO (Extracorporeal Membrane Oxygenation): a life-support technique used when a patient’s heart and/or lungs are failing

Target: Analysize the effective of using ECMO in patients



DATASET

NTUH ECMO dataset

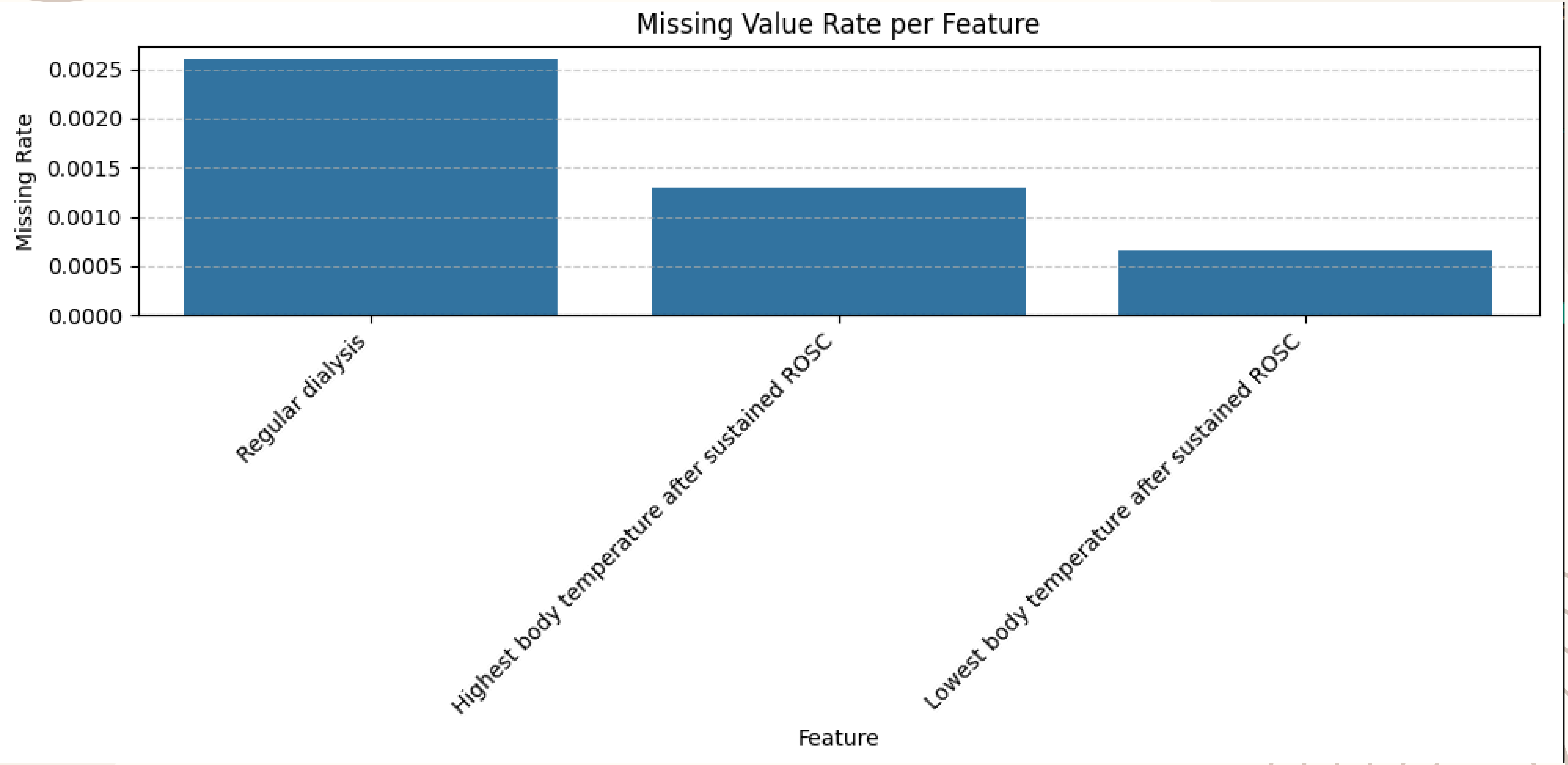
- Link: https://journals.plos.org/plosone/article/file?id=10.1371/journal.pone.0166148&type=printable
- Characteristics:
    - 1,540 patients
    - 47 predictive variables (features)
    - 2 features can be considered as groundtruth:
        - Survival to hospital discharge (1: alive, 0: death)
        - Favorable neurological outcome (1: good, 0: poor)
- We choose target (1: alive, 0: death) for easy classfify

PREPROCESSING DATASET

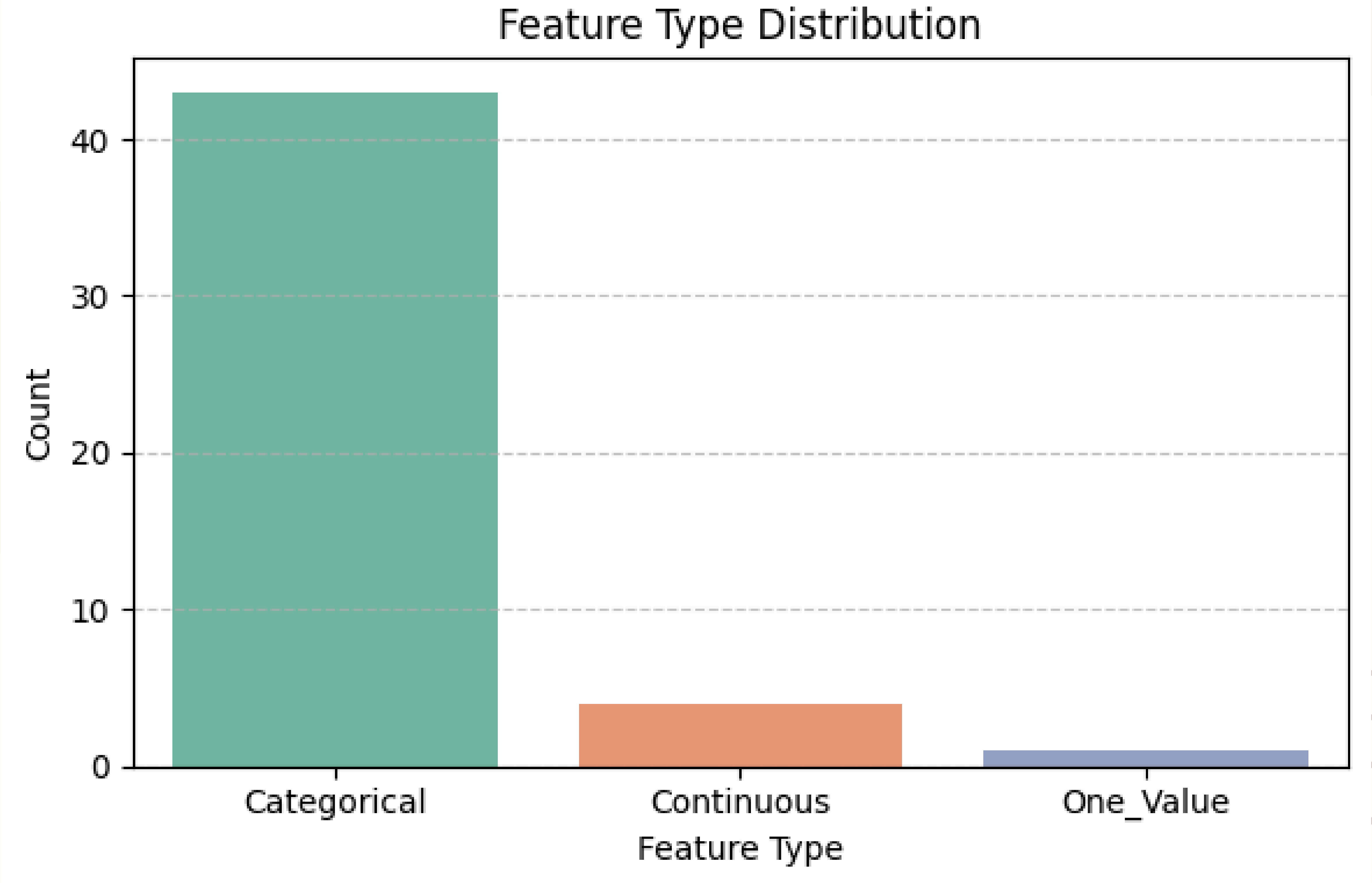
PREPROCESSING

- Cleaning
  - Remove feature “Favorable neurological outcome”
  - Detect NaN values in “Survival to hospital discharge” and delete its row
  - If NaN exists in:
    - Categorical feature(<10): Filling with mode value
    - Continuous feature(>=10): Filling with mean value
    - Unique feature(=1): Remove these rows

ANALYSIZE


Missing Value Rate per Feature
Missing Rate
0.0025
0.0020
0.0015
0.0010
0.0005
0.0000
Regular dialysis
Highest body temperature after sustained ROSC
Lowest body temperature after sustained ROSC
Feature

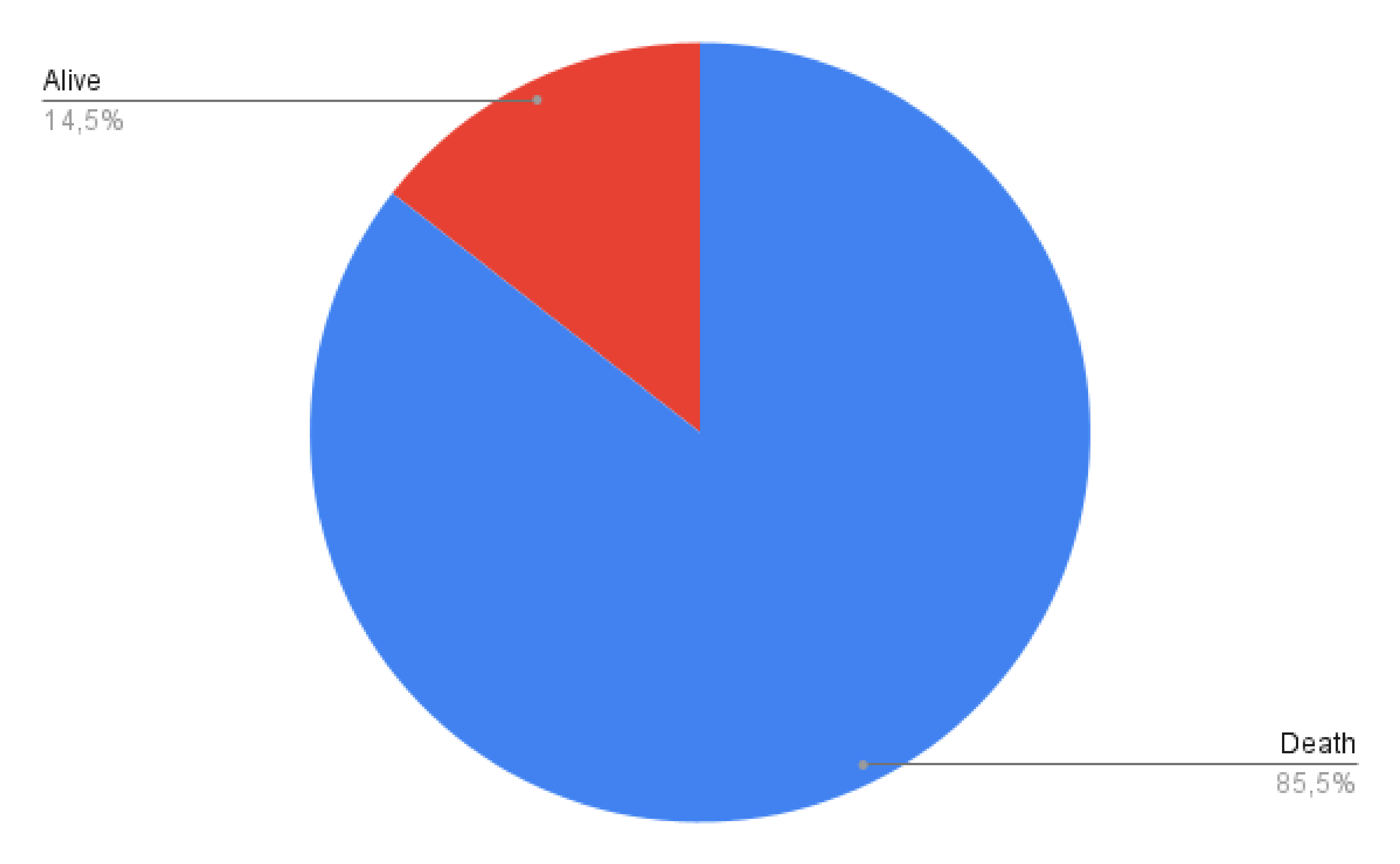
ANALYSIZE


Feature Type Distribution
Count
0
10
20
30
40
Categorical
Continuous
One_Value
Feature Type

ANALYSIZE

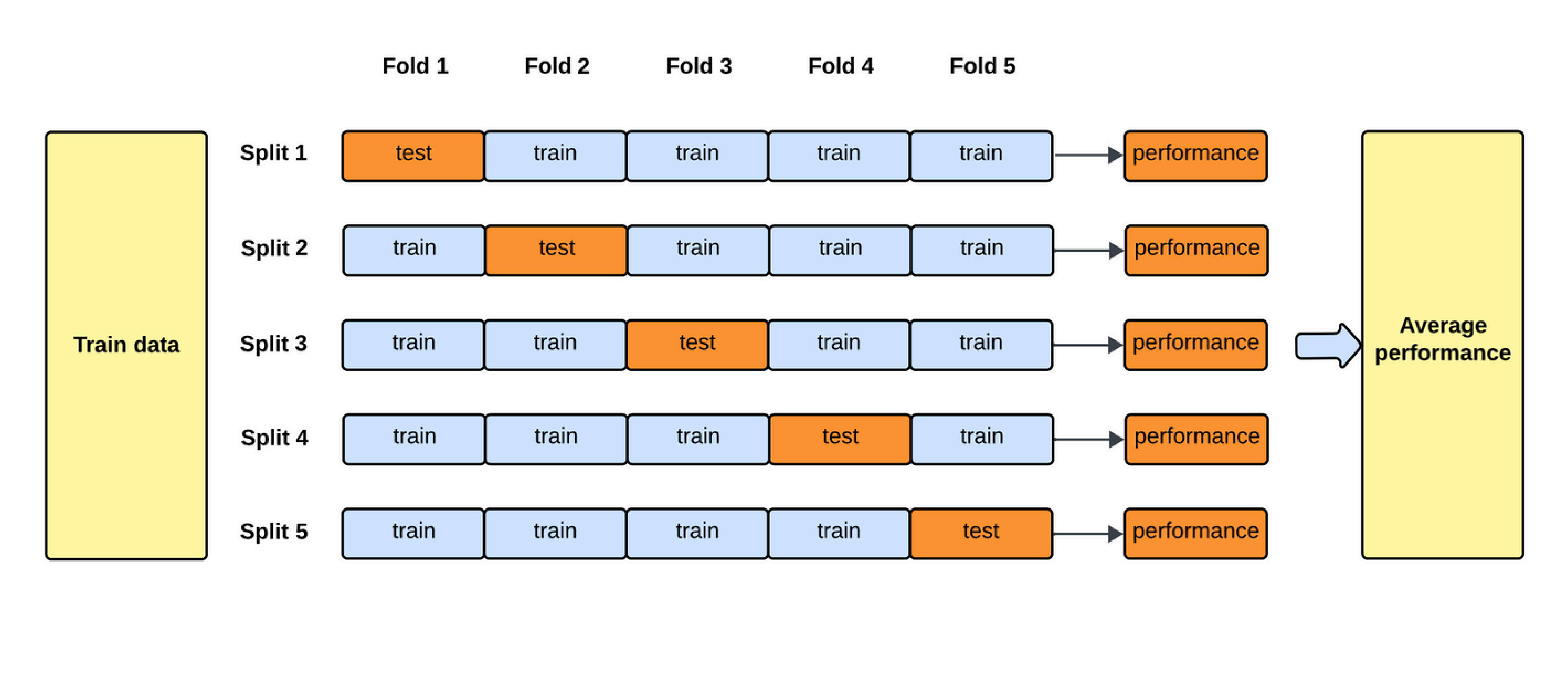
- Groundtruth (Label)
  - Alive: 223 (14.5%)
  - Death: 1317 (85.5%)

→ Imbalanced dataset



DATA SPLITTING

- Using 5-fold cross-validation with:
  - Training: 80%
  - Validation: 10%
  - Testing: 10%

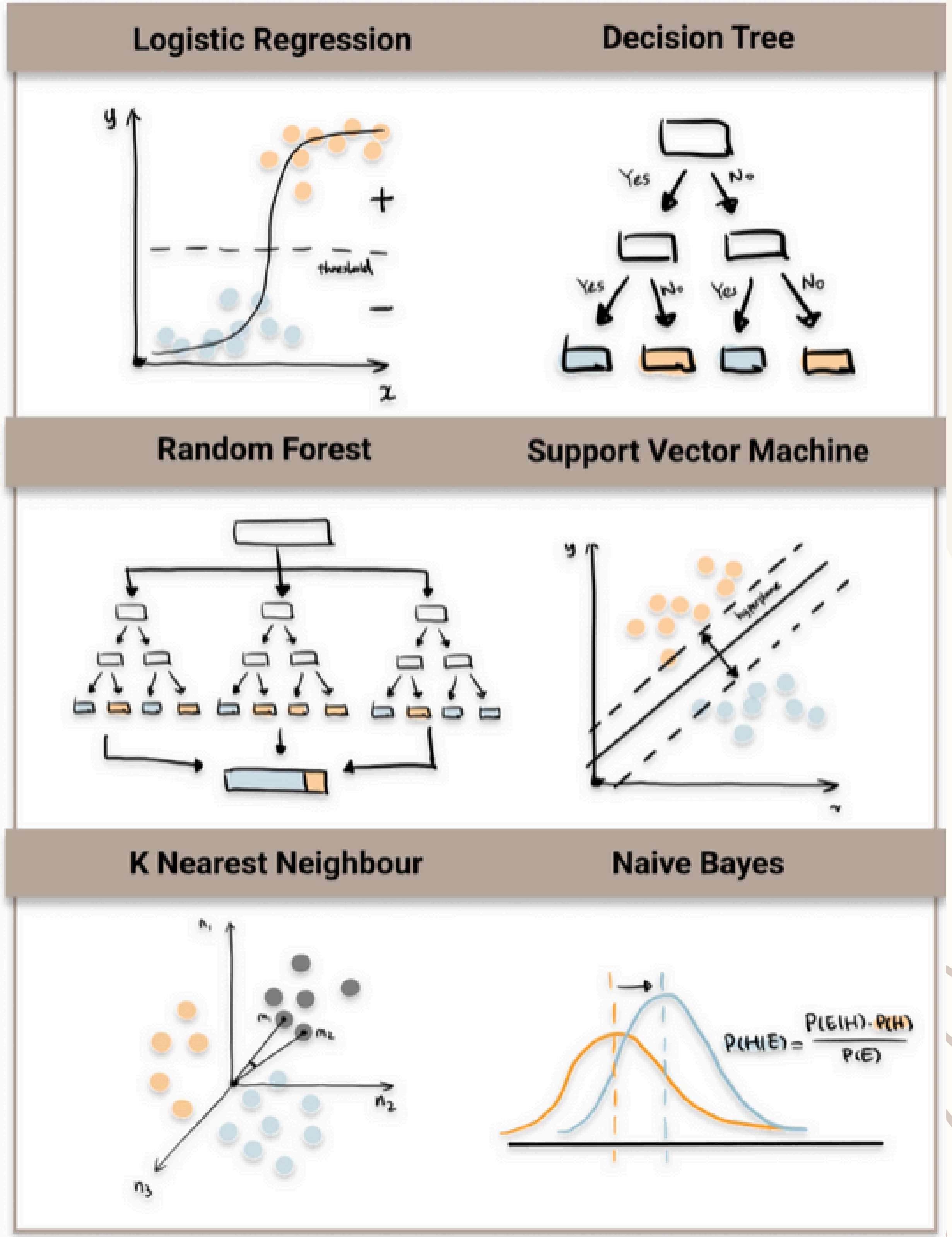


METHODS

MACHINE LEARNING ALGORITHMS

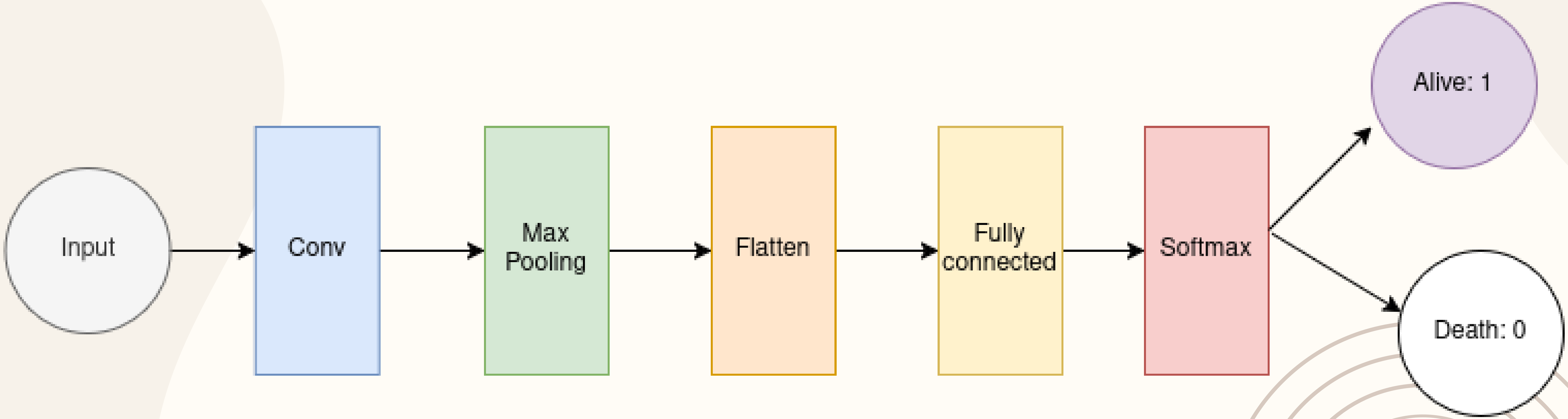
- Decision tree
- Gaussian NB
- KNN
- Logistic regression
- MLP
- Multinomial NB
- Random forest
- XG Boost

→ Suitable for classification problem

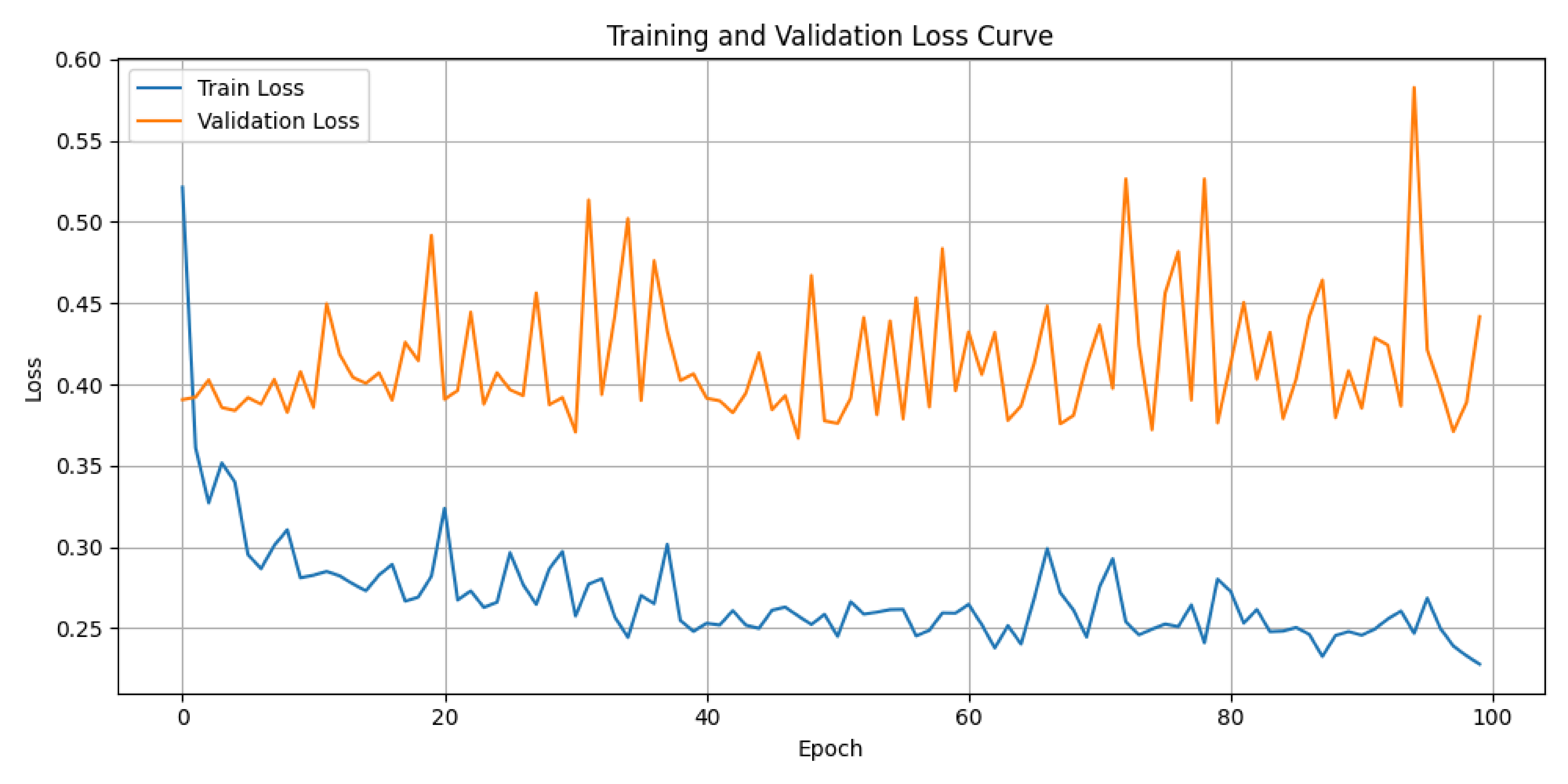


DEEP LEARNING

Simple deep learning with 1 Convolution



TRAINING



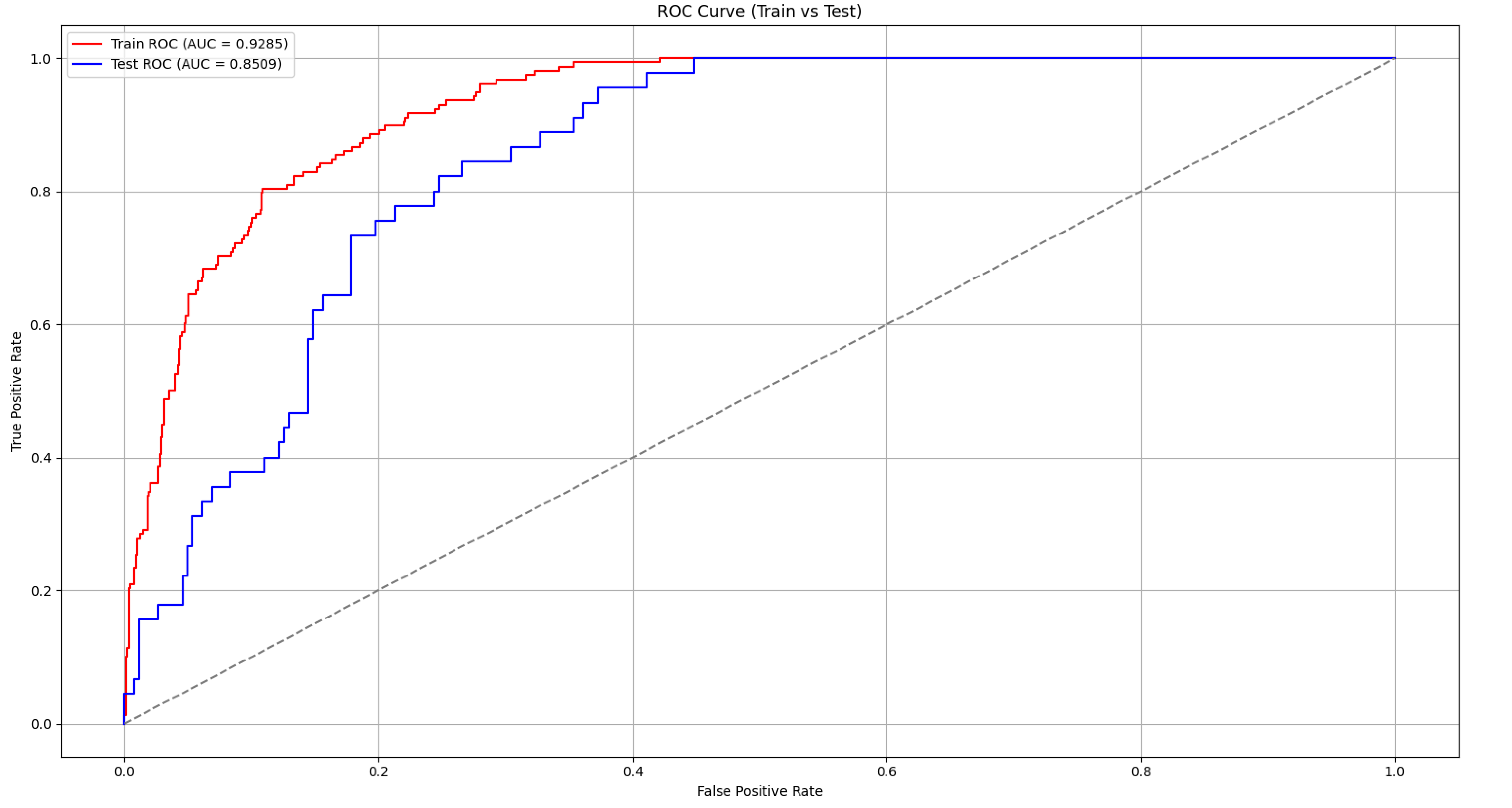
Training and Validation Loss Curve
Train Loss
Validation Loss
Loss
Epoch
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0
20
40
60
80
100

RESULTS

PERFORMANCE

	Accuracy	Sensitivity	Specificity
Decision tree	0.8214	0.4133	0.8912
Gaussian NB	0.6850	0.8755	0.6524
KNN	0.8396	0.2311	0.9437
Logistic regression	**0.8571**	0.3111	**0.9505**
MLP	0.8461	0.3333	0.9338
Multinomial NB	0.6623	0.9244	0.6174
Random forest	**0.8584**	0.2133	**0.9688**
XGBoost	0.8564	0.4133	0.9323
CNN	**0.8513**	0.2267	**0.9582**

AUROC


ROC Curve (Train vs Test)
Train ROC (AUC = 0.9285)
Test ROC (AUC = 0.8509)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

CONCLUSION

CONCLUSION

- Imbalanced dataset is the main cause of low sensitivity
- Almost machine learning algorithms and deep learning can perform well in accuracy
  - Gaussian NB (68.5%) and Multinomial NB (66.23%) are not suitable for this classification problem
  - Logistic Regression (85.71%), Random forest (85.84%) and CNN (85.13%) have high performance

APPENDIX

- Paper:
  - https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0166148
  - https://pubmed.ncbi.nlm.nih.gov/37715216/
- Dataset:
  - https://journals.plos.org/plosone/article/file?id=10.1371/journal.pone.0166148&type=printable
- Code:
  - https://github.com/KhanhLinh-2904/Geriatric-Health
- Tools:
  - Python
  - Excell

Thank You